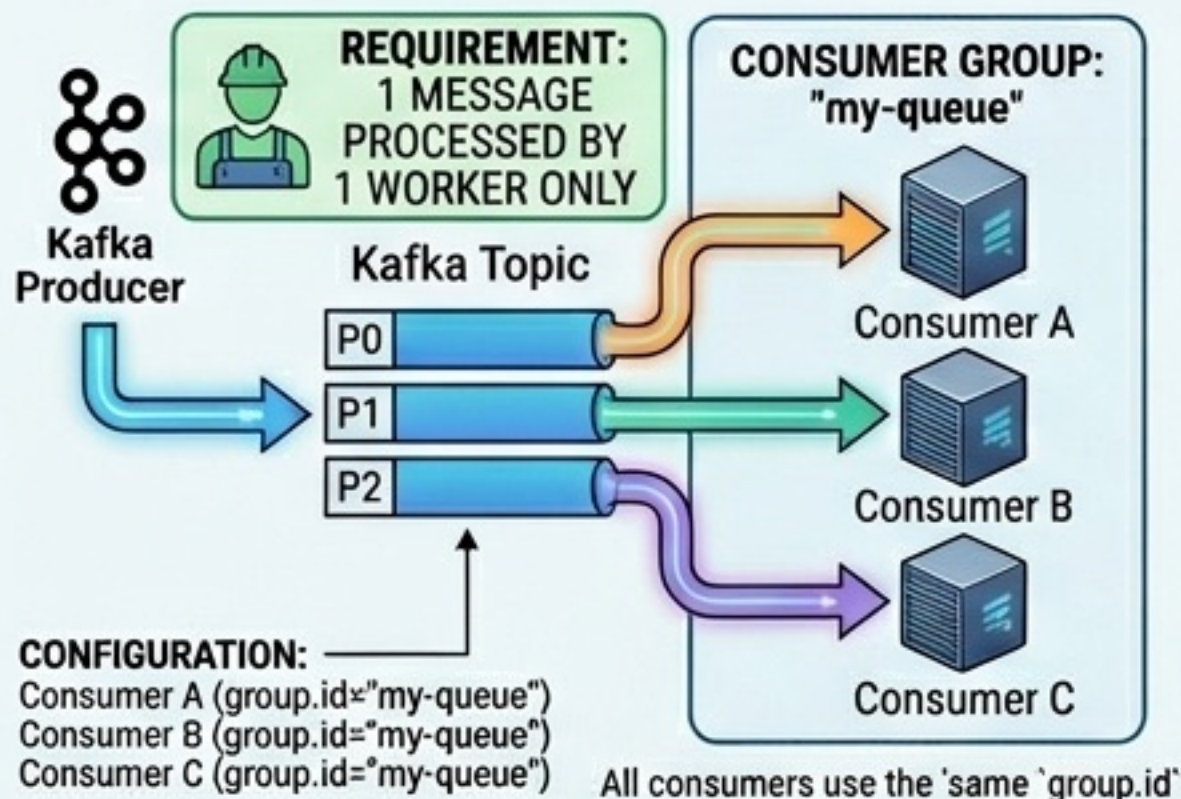


UNDERSTANDING KAFKA'S DUAL PERSONALITY: QUEUE vs. PUB/SUB via `group.id`

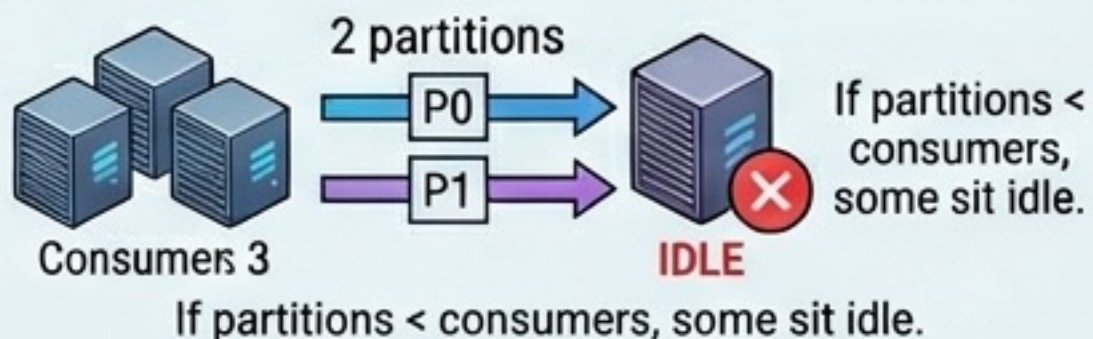
1. ACHIEVING A TRADITIONAL QUEUE (LOAD BALANCING)



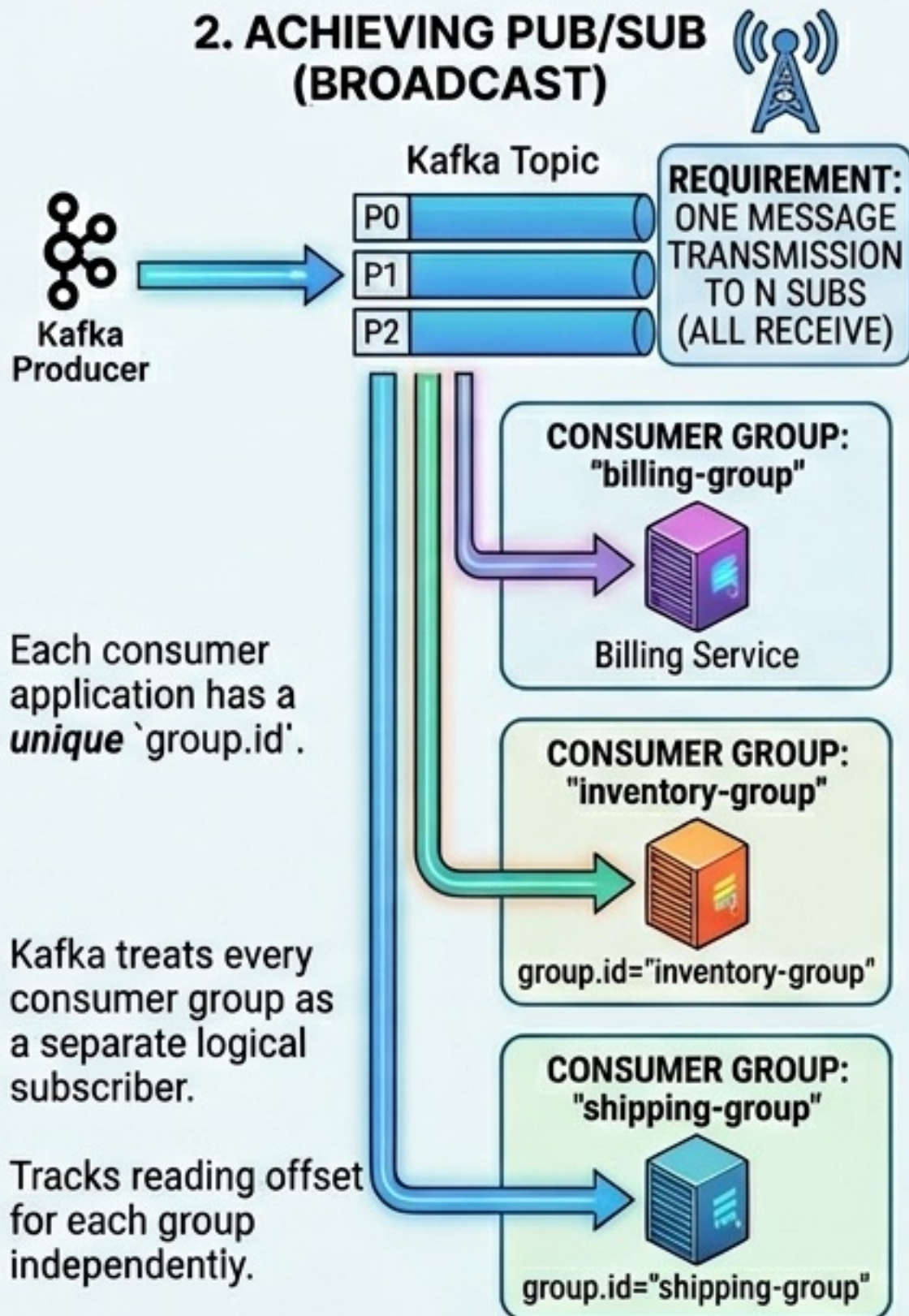
What Kafka does: Kafka divides the topic's partitions exclusively among consumers in the group.

The Result: When a message is produced, it goes to *only one* consumer in the group.

CRUCIAL CAVEAT: CONCURRENCY LIMITED BY PARTITION COUNT



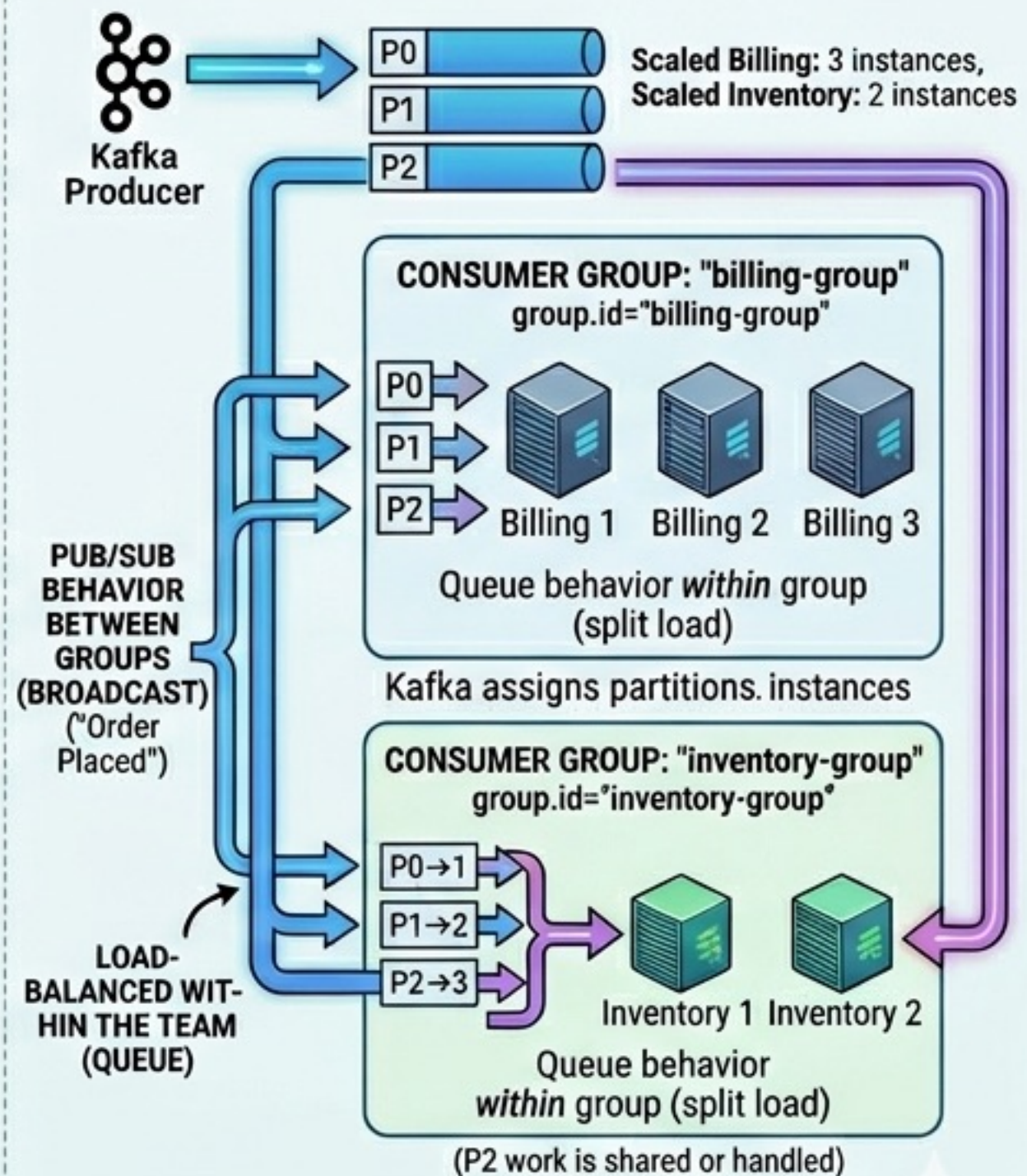
2. ACHIEVING PUB/SUB (BROADCAST)



The Result: One consumer in each group gets a copy of the message.

3. THE HYBRID MODEL: MIXING BOTH SIMULTANEOUSLY

Real-world microservices use this pattern.



Every message is delivered to both Billing and Inventory teams, but within teams, work is load-balanced among instances.